Exercise Sheet 7

Structural Balance

5942UE Network Science

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7.A Triadic Balance Theory

Exercise 7.A.1: Classifying Signed Triangles

Label each signed triangle as **balanced (B)** or **unbalanced (U)**:

1. $A \overset{+}{-} B, B \overset{+}{-} C, A \overset{+}{-} C$ (all positive)
2. $A \overset{+}{-} B, B \overset{+}{-} C, A \overset{-}{-} C$ (two positive, one negative)
3. $A \overset{-}{-} B, B \overset{-}{-} C, A \overset{+}{-} C$ (one positive, two negative)
4. $A \overset{-}{-} B, B \overset{-}{-} C, A \overset{-}{-} C$ (all negative)

Which pattern is most stable, and which is "socially unstable"?

Solution:

A triangle is balanced iff the product of edge signs is positive (0 or 2 negatives).

1. **(+,+,+):** Balanced. Three mutual friends — most stable.
2. **(+,+,−):** Unbalanced. Classic "two of my friends hate each other" tension.
3. **(−,−,+):** Balanced. Two allies sharing a common enemy.
4. **(−,−,−):** Unbalanced. Three mutual enemies — no coalition possible.

The (+,+,−) pattern is the most common source of social tension; (+,+,+) is the most stable.

Exercise 7.A.2: Applying the Balance Theorem

Complete signed graph on $1, 2, 3, 4, 5$. All edges positive **except** 1–3, 1–4, 2–3, 2–4 (which are negative).

1. Check whether every triangle is balanced.
2. If balanced, find the **two-camp partition** guaranteed by the Balance Theorem.
3. Verify within-camp edges are positive and between-camp edges negative.
4. What does the partition represent socially?

Solution:

1. Examples: $1, 2, 3$: (+,−,−) → balanced. $1, 2, 4$: (+,−,−) → balanced. $1, 3, 4$: (−,−,+) → balanced. $3, 4, 5$: (+,+,+) → balanced. All triangles balanced.

2. Partition: $X = 1, 2, 5, Y = 3, 4$.

3. Verification: All within-X edges (1–2, 1–5, 2–5) are positive; the within-Y edge 3–4 is positive; all X–Y edges are negative. ✓

4. Interpretation: Two allied factions in mutual conflict — the textbook two-bloc geopolitical structure ([?EasleyKleinberg2010]).

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7.B Weak Balance

Exercise 7.B.1: Strong vs. Weak Balance

Graph with two triangles connected by one edge:

- A, B, C all positive
- D, E, F all negative
- Connecting edge C–D is negative

1. Is A, B, C balanced under strong balance? Under weak balance?
2. Is D, E, F balanced under strong balance? Under weak balance?
3. What does the global partition look like under each?
4. In international relations, what does an all-negative triangle represent?

Solution:

1. A, B, C — all positive: balanced under both strong and weak balance.

2. D, E, F — all negative: **unbalanced** under strong balance (product is -1). Weak balance *forbids only* (+,+,−) triangles, so all-negative is **balanced** (weakly).

3. Partition:

- *Strong balance:* impossible without resolving D, E, F (one edge must flip).
- *Weak balance:* allows **more than two camps**, e.g. A, B, C, D, E, F — three mutual enemies form three separate factions.

4. Three mutually hostile countries. Strong balance theory predicts an alliance forms (resolving to two camps). Weak balance treats multi-polar rivalry as stable.

Exercise 7.B.2: Finding Camps in a Signed Network

A signed graph: $1, 2, 3$ are mutually friends, $4, 5, 6$ are mutually friends, all cross-group edges are negative.

1. Check every triangle for balance.
2. Find the two-camp partition.
3. Verify within-group edges are positive, between-group negative.
4. Add a "rogue" positive edge $1 \overset{+}{-} 4$. Which triangles become unbalanced?

Solution:

```
import networkx as nx
from itertools import combinations

G = nx.Graph()
G.add_edges_from([(1,2),(1,3),(2,3),(4,5),(4,6),(5,6)], sign=1)
G.add_edges_from([(1,4),(2,5),(3,6),(1,5),(2,6),(3,4)], sign=-1)

def triangle_sign(G, t):
    a, b, c = t
    return G[a][b]["sign"] * G[b][c]["sign"] * G[a][c]["sign"]

triangles = [t for t in combinations(G.nodes, 3)
             if all(G.has_edge(u, v) for u, v in combinations(t, 2))]
balanced = sum(1 for t in triangles if triangle_sign(G, t) > 0)
```

The partition $1, 2, 3$ vs. $4, 5, 6$ makes every triangle balanced. Adding the rogue edge $1 \overset{+}{-} 4$ creates unbalanced triangles such as $1, 2, 4$ and $1, 3, 4$ (each becomes (+,+,−)). The structural tension typically resolves either by breaking the rogue tie or by realigning groups.

7.C Relaxed Balance and Applications

Exercise 7.C.1: WWI Alliance Network

Five countries during WWI:

- **Allies**: France, Britain, Russia (mutually positive)
- **Central Powers**: Germany, Austria-Hungary (mutually positive)
- All Allies–Central Powers edges are negative

1. Is this network perfectly balanced? Identify any unbalanced triangles.
2. Does it match the two-camp Balance Theorem?
3. What if a new country joins with positive ties to **both** camps?
4. Italy switched from Germany to the Allies in 1915 — what does balance theory predict?

Solution:

1. All within-camp triangles are (+,+,+). Cross-camp triangles like F, G, B are (−,−,+) — balanced. The network is perfectly balanced.

2. Yes — exactly the two-camp realisation ([?EasleyKleinberg2010]).

3. A new country with positive ties to both camps creates (+,+,−) triangles immediately — unstable. Balance theory predicts the country must eventually choose a side.

4. Italy initially had a positive tie to Germany while Allies–Germany was negative — straddling two hostile camps. The (+,+,−) tension is exactly what balance theory predicts must resolve, and the historical switch in 1915 eliminated the unbalanced triangles.

Exercise 7.C.2: Approximate Balance in Real Networks

Using the karate club graph, sign edges by faction (within = +, cross = −).

1. Count balanced vs. unbalanced triangles.
2. What fraction of triangles are balanced?
3. Remove cross-faction edges in order of how many unbalanced triangles they participate in. After removing the top 5, how does the fraction change?
4. Plot fraction-balanced vs. edges-removed.

Solution:

```python
import networkx as nx
from itertools import combinations

G = nx.karate_club_graph()
factions = nx.get_node_attributes(G, "club")
for u, v in G.edges():
    G[u][v]["sign"] = 1 if factions[u] == factions[v] else -1

def fraction_balanced(G):
    triangles = [t for t in combinations(G.nodes, 3)
                 if all(G.has_edge(u, v) for u, v in combinations(t, 2))]
    if not triangles:
        return 1.0
    pos = sum(1 for (a, b, c) in triangles
              if G[a][b]["sign"] * G[b][c]["sign"] * G[a][c]["sign"] > 0)
    return pos / len(triangles)
```

Real networks are typically *approximately* balanced (often ≥ 75). Removing high-impact negative edges — those in many unbalanced triangles — improves the global balance fraction sharply, mirroring how social groups resolve tension by severing conflicting ties.

Wrap-Up

- triangle sign products give a one-line balance test
- the Balance Theorem turns a local rule into a global two-camp partition
- weak balance permits more than two camps and is often more realistic
- real networks are rarely perfectly balanced — approximate balance is the typical regime